Fiscal Year:	FY 2015	**Task Last Updated:**	FY 03/13/2015
PI Name:	Datta, Kamal M.D.		
Project Title:	High-LET Radiation, Reproductive Hormones and Chronic Inflammation: An Integrated Approach to Carcinogenesis Risk Estimate		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20057-0001	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	01/01/2013	**End Date:**	12/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2015, per NSSC information (Ed., 12/17/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AD58G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of this project is to improve NASA risk estimates for space radiation-induced intestinal carcinogenesis. Inflammation and oxidative stress is intimately linked and both are strongly implicated in gastrointestinal (GI) carcinogenesis, and exposure to radiation results in production of reactive oxygen species (ROS) leading to oxidative stress. High levels of ROS are known to induce an apoptotic response but sustained sub-lethal oxidative stress has the potential to produce persistent modification of biomolecules including proteins, DNA, and lipids leading to alterations in their functions with adverse cellular consequences. Exposure of cells to stress is known to activate mitogenic signaling pathways such as EGFR/RAS/PI3KMAPK-related, which in turn stimulates proliferation and progression into mitosis. Persistent proliferative signals in the presence of DNA damage have been projected to propagate mutation in the progeny with survival advantage and carcinogenesis. Chronic sub-lethal oxidative stress could activate stress-sensitive kinases and transcription factors such as NFkB and AP1, and cause increased production of inflammatory cytokines. Although long-term effects of low linear energy transfer (low-LET) radiation such as gamma radiation on persistent oxidative stress and inflammation and how they relate to GI carcinogenesis have been investigated, information about the influence of high-LET cosmic radiation on GI cancer is scarce. Therefore, the overall goal of this proposal is to understand the effects of cosmic radiation on persistent oxidative stress and inflammation in the GI tract and how these alterations relate to GI carcinogenesis. In conjunction with our GI NASA Specialized Center of Research (NSCOR) program, the current proposal aims to compare the results of highly energetic charged (HZE) particle radiation to gamma radiation in mice to improve NASA's carcinogenesis risk estimates for GI cancer in male and female astronauts through a mechanistic understanding of the underlying biological processes. The specific aim for the proposal is therefore to characterize the roles of HZE radiation-induced persistent oxidative stress and inflammatory responses in gastrointestinal carcinogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S., and American Cancer Society estimates that there will be 132,700 new cases of colorectal cancer (CRC) with 49,700 persons predicted to die of the disease in 2015. Colorectal cancer when considered in men and women is the second leading cancer killer in the United States (http://www.cdc.gov/). Considering the high frequency of colorectal tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. Inflammation and cellular stress are critical elements of carcinogenesis and both are associated with alterations in biomolecules such as DNA. Because radiation is also known to initiate cellular oxidative stress and inflammatory responses, HZE radiation due to its qualitative difference from terrestrial radiation is expected to induce long-term stress and inflammatory responses thus increasing the risk of GI cancer.
Task Progress:	Persistent oxidative stress has been implicated in gastrointestinal (GI) carcinogenesis. The overall aim is to characterize the effects of space radiation in triggering persistent stress responses and their potential effects on tumorigenesis in mouse intestine. This will involve quantitative and qualitative measurements of long-term cellular oxidative stress and related inflammatory responses in serum and in intestinal tissues after heavy ion and gamma-radiation. While we are continuing our investigation on oxidative stress and related inflammatory responses in serum and in intestinal tissues, we have also focused our attention on intestinal stem cells. We have carried out three beam runs in the second year of this project at the NASA Space Radiation Laboratory (NSRL) at Brookhaven National Laboratory (BNL). We standardized intestinal stem cell isolation protocol from GFP expressing Lgr5+ (Lgr5-EGFP-IRES-creERT2) mice exposed to sham and different types of radiation using FACS-based cell sorting. Sorted stem cells were used to measure reactive oxygen species and assess mitochondrial status two months after exposure. Additionally, intestinal sections were stained for gamma H2AX, which showed persistent DNA damage even two months after 56-Fe radiation. We exposed wild type C57BL/6J mice to different types of radiation to define how heavy ion radiation exposure affects the epithelial cell migration, which is important for maintenance of intestinal homeostasis and is perturbed in intestinal pathologies including cancer. Exposed mice were intraperitoneally injected with BrdU seven days after radiation and mice were sacrificed 4, 24, and 48 h after injection. Intestinal sections were stained for anti-BrdU and analysis showed altered cell migration after irradiation. Currently, work on molecular pathways involved in intestinal epithelial cell migration is ongoing and works on two-month post-exposure samples are also in progress.
Bibliography Type:	Description: (Last Updated: 06/25/2025)
Abstracts for Journals and Proceedings	Cheema AK, Suman S, Kaur P, Singh R, Fornace AJ Jr, Datta K. "Longterm differential changes in mouse intestinal metabolomics after gamma and heavy ion radiation exposure." 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. Abstract Book. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. , Sep-2014
Articles in Peer-reviewed Journals	Cheema AK, Suman S, Kaur P, Singh R, Fornace AJ Jr, Datta K. "Long-term differential changes in mouse intestinal metabolomics after gamma and heavy ion radiation exposure." PLoS One. 2014 Jan 27;9(1):e87079. eCollection 2014. http://dx.doi.org/10.1371/journal.pone.0087079 ; PubMed PMID: 24475228; PubMed Central PMCID: PMC3903607 , Jan-2014
Articles in Peer-reviewed Journals	Suman S, Kallakury BV, Fornace AJ Jr, Datta K. "Protracted upregulation of leptin and IGF1 is associated with activation of PI3K/Akt and JAK2 pathway in mouse intestine after ionizing radiation exposure." Int J Biol Sci. 2015 Jan 20;11(3):274-83. eCollection 2015. http://dx.doi.org/10.7150/ijbs.10684 ; PubMed PMID: 25678846; PubMed Central PMCID: PMC4323367 , Jan-2015
Articles in Peer-reviewed Journals	Datta K, Suman S, Fornace AJ Jr. "Radiation persistently promoted oxidative stress, activated mTOR via PI3K/Akt, and downregulated autophagy pathway in mouse intestine." Int J Biochem Cell Biol. 2014 Dec;57:167-76. Epub 2014 Oct 23. http://dx.doi.org/10.1016/j.biocel.2014.10.022 ; PubMed PMID: 25449263 , Dec-2014
Articles in Peer-reviewed Journals	Suman S, Moon BH, Thakor H, Fornace AJ Jr, Datta K. "Wip1 abrogation decreases intestinal tumor frequency in APC(Min/+) mice irrespective of radiation quality." Radiat Res. 2014 Sep;182(3):345-9. Epub 2014 Aug 12. http://dx.doi.org/10.1667/RR13770.1 ; PubMed PMID: 25117622 , Sep-2014